

Supplementary Information for Election-Denying Republican Candidates Underperformed in the 2022 Midterms

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May 28, 2024

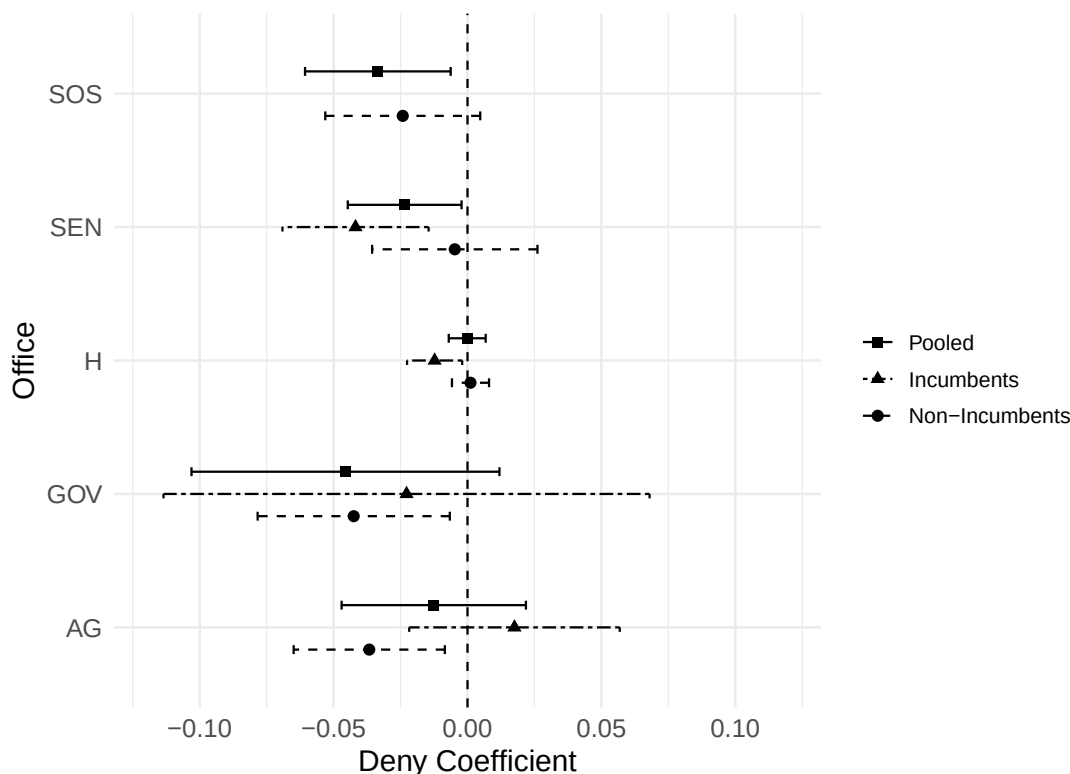
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A Incumbency and the Election-Denying Candidate Underperformance Across Offices

There are two important potential sources of variation in how much election-denying candidates underperform. First, denying the election might be more important for political actors with direct influence over the ways state elections function; hence, we might expect the issue to be more salient and underperformance to be larger in races for secretary of state, governor, and attorney general. On the other hand, members of the House do vote on the certification of the Electoral College, so who voters will think of as most responsible for election administration is unclear. Second, the other advantages of incumbency might serve to dampen the underperformance, allowing incumbents more freedom to deny the election if they so wish. Figure SI.1 explores both of these patterns. In the figure, we plot coefficient estimates that reflect the overall estimate for each office, as well as the underperformance separately for incumbents and non-incumbents.

Figure SI.1 – The Underperformance of Election-Denying Candidates Across Offices and Incumbency.



Note: No Secretary of State incumbents denied the election results in our sample; hence, this estimate is omitted from the top row of the plot.

Although we are cutting a relatively small dataset quite finely, there is some evidence that the underperformance is larger in statewide offices—specifically in secretary of state and governor races. The underperformance to AGs seems smaller, though, and the underperformance to US Senators seems larger as well. The House stands out as the office where the underperformance is smallest (and indeed, zero). We formally test whether the underperformance to statewide offices and the underperformance to Congress are significantly different and find that the underperformance to statewide offices is 3.5 percentage points larger than the House and Senate with a p-value of 0.006.

Table SI.1 – Possible Advantage of Election-Denying Candidates in 2022 Republican Primary Races for Governor, Secretary of State, and Attorney General.

	Primary-Election Vote Share		
	(1)	(2)	(3)
Election-Denying Candidate	0.017 (0.027)	0.019 (0.031)	0.016 (0.036)
N	293	291	277
# States	36	34	33
Pres Vote Share	Yes	No	No
# Cand FEs	Yes	Yes	Yes
State FE	No	Yes	No
State by Office FE	No	No	Yes

Robust standard errors clustered by state in parentheses.
All columns use States United classification for election deniers, which is the only classification that includes all primary candidates.

The explanation of these discrepancies is unclear and will require follow-up research after the 2024 election cycle with more data.

Interestingly, we do not find a consistent pattern with respect to incumbency. No secretary of state incumbents denied the election, so the entire underperformance estimated in these offices accrues to non-incumbents. In governor and AG races, the underperformance appears to be larger for non-incumbents than for incumbents. On the other hand, in the House and Senate, the opposite appears to be the case; here, we document underperformance for incumbents but not for non-incumbents.

B Advantage to Election-Denying Candidates in Primary Elections

While election-denying candidates may be disadvantaged in the general election, many observers have concluded that they enjoyed a considerable advantage in primary elections. To investigate this, we can compute the average vote-share advantage or disadvantage for election-denying candidates. We run regressions of the form

$$\text{Primary Vote Share}_{ios} = \beta \text{Deny 2020}_i + \delta X_{ios} + \epsilon_{ios}, \quad (1)$$

where the outcome variable is candidate i 's primary-election vote share for office o in state s . The main explanatory variable is the same as before. Here, optional controls will include presidential vote share, state fixed effects, or state-by-office fixed effects. We also control in all specifications for the number of candidates in the race by using fixed effects for each possible number of candidates observed in the data. This is important because each additional candidate in a primary mechanically (weakly) reduces the average vote share of all candidates in the race.

Table SI.1 presents the results for these three specifications. Across the specifications, we see a relatively stable but very noisy estimate ranging from 1.6 percentage points in column 1 to

1.9 percentage points in column 2. In all three specifications, standard errors are large, and we cannot rule out somewhat large effects. This important limitation notwithstanding, the estimated advantage seems surprisingly small relative to some prominent claims.

C Distribution of Election-Denying Candidates

In this section, we plot the number of denying and non-denying candidates by office and incumbency status, using our combined measure of election denialism.

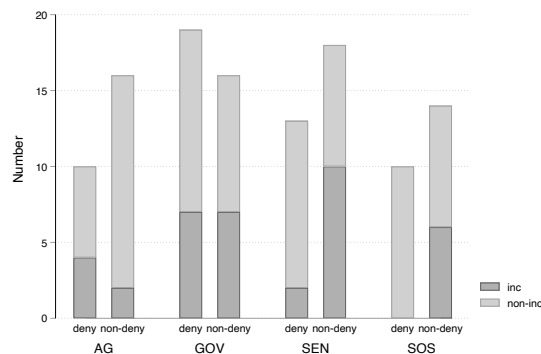


Figure SI.2 – Barplot plots the number of election-denying and non-denying candidates across office and incumbency status.

D Alternative 538 Classification

In this section, we re-estimate our main results with a different classification of election-denying candidates. Here, using the 538 data, we define election-denying candidates as anyone who either “fully denied” the election results or “raised questions” about it. The results are presented in Table SI.2.

Table SI.2 – 538 Denier Classification Robustness of Underperformance of Election-Denying Candidates in 2022 Races for Governor, Senator, Secretary of State, and Attorney General.

	General-Election Vote Share							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Election-Denying Candidate	-0.026 (0.010)	-0.019 (0.010)	-0.026 (0.009)	-0.028 (0.011)	-0.020 (0.012)	-0.022 (0.015)	-0.025 (0.012)	-0.026 (0.018)
N	116	116	116	95	53	69	68	42
# States	42	42	42	39	16	20	19	13
Pres Vote Share	Yes	Yes	Yes	Yes	No	No	No	No
State FEs	No	No	No	No	Yes	Yes	Yes	Yes

Sample is Republican candidates for attorney general, secretary of state, governor, and U.S. Senate. Robust standard errors clustered by state in parentheses. Columns (1) and (5) use the 538 denier classification from the paper that only classifies a candidate as an election denier for the stance 'fully denied'. Columns (2) and (6) classify a candidate as a denier if their 538 stance was 'fully denied', 'no comment', or 'avoided answering'. Columns (3) and (7) classify a candidate as a denier if their 538 stance was 'fully denied' or 'raised questions'. Columns (4) and (8) use the same 538 classification from the paper except omit observations with the stances 'avoided answering' and 'no comment'. Columns 1-4 include controls for 2020 presidential voteshare.

Table SI.3 – Underperformance of Election-Denying Candidates in 2022 Races for Governor, Senator, Secretary of State, and Attorney General.

	General-Election Vote Share							
	(1) States United	(2) 538	(3) WaPo	(4) Combined	(5) States United	(6) 538	(7) WaPo	(8) Combined
Election-Denying Candidate	-0.033 (0.015)	-0.026 (0.010)	-0.030 (0.010)	-0.029 (0.010)	-0.045 (0.026)	-0.020 (0.012)	-0.028 (0.013)	-0.032 (0.012)
Dem Pres Voteshare	-0.921 (0.091)	-0.967 (0.068)	-0.962 (0.066)	-0.971 (0.066)				
N	85	116	116	116	32	53	72	68
# States	38	42	42	42	11	16	21	20
Pres Vote Share	Yes	Yes	Yes	Yes	No	No	No	No
State FEs	No	No	No	No	Yes	Yes	Yes	Yes

Sample is Republican candidates for attorney general, secretary of state, governor, and U.S. Senate. Robust standard errors clustered by state in parentheses. Columns labeled States United use election-denier classifications from SUDC, which do not include Senate or House races. Columns labeled 538 use election denier classifications from 538. Rows labeled WaPo use election denier classifications from the Washington Post. Columns 1-4 include controls for 2020 presidential voteshare

E Full Table Specifications

This section provides fuller versions of the general-election (Table SI.3) and primary-election regression (Table SI.4) tables showing the coefficients of the control variable for presidential vote share.

Table SI.4 – Full Version of Possible Advantage of Election-Denying Candidates in 2022 Republican Primary Races for Governor, Senator, Secretary of State, and Attorney General.

	Primary-Election Vote Share		
	(1)	(2)	(3)
Election-Denying Candidate	0.017 (0.027)	0.019 (0.031)	0.016 (0.036)
Dem Pres Voteshare	0.043 (0.047)		
N	293	291	277
# States	36	34	33
Pres Vote Share	Yes	No	No
# Cand FEs	Yes	Yes	Yes
State FE	No	Yes	No
State by Office FE	No	No	Yes

Robust standard errors clustered by state in parentheses.
All columns use States United classification for election deniers, which is the only classification that includes all primary candidates.

F List of Election-Denying Candidates

In this section, we show the raw data on which statewide candidates are classified as election-denying candidates across our three data sources. The results can be found in Tables SI.5-SI.13.

A ✓ indicates that the candidate was classified as denying the results of 2020 election by either 538, WaPo, or States United. A blank indicates the candidate was classified as not having rejected the election. A "-" indicates that the candidate was not considered by the source.

Table SI.5 – Republican Candidates for Attorney General Classified as Deniers

State	Candidate	General Election	538	SUDC	WaPo
AL	MARSHALL, STEVE	Won		✓	✓
AZ	GLASSMAN, RODNEY		-	✓	-
AZ	HAMADEH, ABRAHAM	Lost	✓	✓	✓
CA	EARLY, ERIC		-	✓	-
FL	MOODY, ASHLEY	Won	✓	✓	✓
GA	GORDON, JOHN		-	✓	-
ID	LABRADOR, RAÚL R.	Won	✓	✓	
ID	MACOMBER, ARTHUR		-	✓	-
IL	SHESTOKAS, DAVID		-	✓	-
KS	KOBACH, KRIS	Won	✓	✓	✓
MD	PEROUTKA, MICHAEL ANTHONY	Lost		✓	✓
MI	DEPERNO, MATTHEW	Lost	✓	✓	✓
MN	ANDERSON, SHARON		-	✓	-
MN	WARDLOW, DOUG		-	✓	-
NV	CHATTAH, SIGAL	Lost		✓	✓
OH	YOST, DAVE	Won			✓
SC	WILSON, ALAN	Won	✓	✓	✓
TX	GOHMERT, LOUIE		-	✓	-
TX	PAXTON, KEN	Won	✓	✓	✓
VT	PAIGE, H. BROOKE		-	✓	-
WI	MUELLER, KAREN		-	✓	-

Table SI.7 – Republican Candidates for Governor Classified as Deniers

State	Candidate	General Election	538	SUDC	WaPo
AL	BLANCHARD, LINDY		-	✓	-
AL	IVEY, KAY	Won	✓	✓	✓
AL	ODLE, DEAN		-	✓	-
AR	WASHBURN, FRANCIS		-	✓	-
AZ	LAKE, KARI	Lost	✓	✓	✓
AZ	NEELY, SCOTT DAVID		-	✓	-
AZ	SALMON, MATT		-	✓	-
AZ	ZEN, PAOLA TULLIANI		-	✓	-
CA	MERCURI, DANIEL R.		-	✓	-
CA	WILLIAMS, MAJOR		-	✓	-
CA	ZACKY, LEO S.		-	✓	-
FL	DESANTIS, RON	Won		✓	✓
GA	PERDUE, DAVID A.		-	✓	-
GA	TAYLOR, KANDISS		-	✓	-
IA	REYNOLDS, KIM	Won		✓	✓
ID	HUMPHREYS, EDWARD R.		-	✓	-
ID	LITTLE, BRAD	Won	✓		✓
ID	MCGEACHIN, JANICE		-	✓	-
IL	BAILEY, DARREN	Lost			✓
IL	SOLOMON, MAX		-	✓	-
KS	SCHMIDT, DEREK	Lost		✓	✓
MA	DIEHL, GEOFF	Lost	✓	✓	✓
MD	COX, DAN	Lost	✓	✓	✓
ME	LEPAGE, PAUL R.	Lost		✓	✓
MI	DIXON, TUDOR M.	Lost	✓	✓	✓
MI	KELLEY, RYAN D.		-	✓	-
MI	REBANDT, RALPH		-	✓	-
MI	SOLDANO, GARRETT		-	✓	-
NE	CONNELY, MICHAEL		-	✓	-
NE	HERBSTER, CHARLES W.		-	✓	-
NE	RIDENOUR, BRELAND		-	✓	-

Table SI.9 – Republican Candidates for Governor Classified as Deniers

State	Candidate	General Election	538	SUDC	WaPo
CO	GANAHL, HEIDI	Lost			✓
MN	JENSEN, SCOTT	Lost			✓
NV	GILBERT, JOEY		-	✓	-
NV	HAMILTON, EDDIE		-	✓	-
NV	LEE, JOHN J.		-	✓	-
NY	GIULIANI, ANDREW		-	✓	-
NY	ZELDIN, LEE	Lost		✓	✓
OH	BLYSTONE, JOE		-	✓	-
OH	HOOD, RON		-	✓	-
OH	RENACCI, JIM		-	✓	-
OK	SHERWOOD, MARK		-	✓	-
OR	CHRISTENSEN, REED		-	✓	-
OR	RICHARDSON, AMBER R.		-	✓	-
PA	BARLETTA, LOUIS		-	✓	-
PA	GALE, JOSEPH CHARLES		-	✓	-
PA	GEROW, CHARLES R.		-	✓	-
PA	MASTRIANO, DOUGLAS V.	Lost	✓	✓	✓
SD	NOEM, KRISTI	Won			✓
TN	LEE, BILL	Won		✓	
TX	ABBOTT, GREG	Won		✓	✓
TX	HARRISON, DANNY		-	✓	-
TX	HUFFINES, DON		-	✓	-
TX	PRATHER, CHAD		-	✓	-
TX	WEST, ALLEN B.		-	✓	-
WI	FISCHER, ADAM J.		-	✓	-
WI	KLEEFISCH, REBECCA		-	✓	-
WI	MICHELS, TIM	Lost		✓	✓
WI	RAMTHUN, TIMOTHY		-	✓	-

Table SI.11 – Republican Candidates for U.S. Senator Classified as Deniers

State	Candidate	General Election	538	SUDC	WaPo
AL	BRITT, KATIE	Won	✓	-	✓
AZ	MASTERS, BLAKE	Lost		-	✓
GA	WALKER, HERSCHEL JUNIOR	Lost		-	✓
KY	PAUL, RAND	Won		-	✓
MD	CHAFFEE, CHRIS	Lost	✓	-	✓
MO	SCHMITT, ERIC	Won	✓	-	✓
NC	BUDD, TED	Won		-	✓
NH	BOLDUC, DONALD C.	Lost		-	✓
NV	LAXALT, ADAM PAUL	Lost	✓	-	✓
OH	VANCE, J. D.	Won	✓	-	✓
OR	PERKINS, JO RAE	Lost	✓	-	✓
PA	OZ, MEHMET C.	Lost		-	✓
WI	JOHNSON, RON	Won	✓	-	

Table SI.13 – Republican Candidates for Secretary of State Classified as Deniers

State	Candidate	General Election	538	SUDC	WaPo
AL	ALLEN, WES	Won		✓	
AR	WILLIAMS, EDDIE JOE		-	✓	-
AZ	BOLICK, SHAWNNA		-	✓	-
AZ	FINCHEM, MARK	Lost	✓	✓	✓
CA	HAMM, RACHEL		-	✓	-
CA	PAINE, JAMES		-	✓	-
CO	PETERS, TINA		-	✓	-
CT	RAPINI, DOMINIC	Lost		✓	✓
GA	BELLE ISLE, DAVID C.		-	✓	-
GA	HICE, JODY		-	✓	-
ID	MOON, DOROTHY		-	✓	-
ID	SOUZA, MARY		-	✓	-
IN	MORALES, DIEGO	Won		✓	✓
KS	BROWN, MIKE		-	✓	-
MA	CAMPBELL, RAYLA	Lost		✓	✓
MI	KARAMO, KRISTINA ELAINE	Lost	✓	✓	✓
MN	CROCKETT, KIM	Lost	✓	✓	✓
MN	VAN MECHELEN, ERIK		-	✓	-
NE	SCHRODER, REX		-	✓	-
NM	TRUJILLO, AUDREY	Lost	✓	✓	✓
NV	GERHARDT, JOHN CARDIFF		-	✓	-
NV	MARCHANT, JIM	Lost	✓	✓	✓
OH	ADAMS, JOHN		-	✓	-
SC	BLANDFORD, KEITH		-	✓	-
VT	PAIGE, H. BROOKE	Lost	✓	✓	✓
WI	SCHMIDTKA, JUSTIN D.		-	✓	-
WI	SCHROEDER, JAY		-	✓	-
WY	GRAY, CHUCK	Won	✓	✓	✓

Table SI.15 – Underperformance of Election-Denying Candidates in 2022 Races by Office Interacted with Incumbency.

	General-Election Vote Share				
	(1)	(2)	(3)	(4)	(5)
	AG	SOS	GOV	SEN	House
deny	-0.037 (0.014)	-0.024 (0.014)	-0.042 (0.018)	-0.005 (0.014)	0.001 (0.003)
inc	0.031 (0.021)	0.025 (0.012)	0.114 (0.059)	0.032 (0.021)	0.035 (0.006)
inc $\times$ deny	0.054 (0.024)		0.020 (0.047)	-0.037 (0.024)	-0.013 (0.005)
N	26	24	35	31	393
Pres Vote Share	Yes	Yes	Yes	Yes	Yes

Sample for each row is Republican candidates for attorney general, secretary of state, governor, U.S. Senate, and U.S. House respectively. Each estimate is from a separate model with only candidates for the given office. Robust standard errors clustered by state in parentheses. All columns use the aggregated denier classification. Columns 1-5 include controls for 2020 presidential voteshare

G Incumbency and the underperformance to Election-Denying Candidates Across Offices

Here we provide Table SI.15 to match the incumbency coefficient plot.